

Aaryan Lath

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Athletic, agile complex systems designer who thrives in team-based environments, using systems-level thinking to translate concepts into hardware/software solutions through rapid prototyping and applied research.

EDUCATION

Purdue University

Bachelor of Science in Aeronautical and Astronautical Engineering (GPA: 3.73/4.0)

West Lafayette, IN

Expected May 2026

Specialization in Systems Design and Aerodynamics

Honors: Warren G. Koerner Scholar, Dean's List and Semester Honors for **every** semester, **AAE Ambassador**

SKILLS

Core Tools: Technical Writing, CAD, Assembly, CREO, NX, GD&T, Teamcenter, Manufacturing, Ansys, SolidWorks, JIRA

Programming & Analysis: System Integration, GenAI Prompting, Python, MATLAB, MS Office, Git, CFD, Simulink

Soft-skills: Analytical, Communicator, Fast-paced learner, Inquisitive, Business Strategy

PROFESSIONAL EXPERIENCE

Siemens USA

Mechanical Engineering Intern

Grand Prairie, TX

May 2025 – Aug 2025

- Optimized panelboard enclosures using **CREO** and executed ECNs in **SAP**, ensuring 100% compliance.
- Streamlined **100+** switchboard assembly configurations, resolving edge cases to reduce design cycle time.
- Automated BOM generation via Python using **AI prompting**, increasing efficiency by **30%** for **10+** daily orders.

Purdue School of Aeronautics and Astronautics

Undergraduate Teaching Assistant – Aerospace System Design

West Lafayette, IN

Jan 2026 – Present

- Automated team selection via HTML and backend optimization, improving course logistics and data flow.
- Graded industry-mimicked SDRs/SRRs for **7** teams, delivering critical feedback on technical strategy.

Undergraduate Teaching Assistant – Intro to Aircraft and Spacecraft Design

Jan 2025 – Dec 2025

- Led study sessions for **50+** students, communicating core principles of Aircraft/Spacecraft design.
- Provided personalized technical support to students with concepts and **MATLAB** troubleshooting.

Resilient Extraterrestrial Habitat Engineering

Undergraduate Research Assistant - Systems Engineer

West Lafayette, IN

May 2024 – May 2025

- Modeled 18 potential disruptions scenarios via Simulink to enhance habitat resilience and risk mitigation.
- Executed trade studies using **Equivalent System Mass** analysis to optimize safety-cost ratios for systems.

PROJECTS AND INVOLVEMENTS

Next-Gen Carrier-Based Military Aircraft Design

Project Engineer

West Lafayette, IN

Jan 2025 – Present

- Generated an aircraft configuration to submit as an **RFP** to **AIAA** for Aircraft Design Competition.
- Selected aircraft powerplant and performed inlet sizing on **MATLAB** based on my selected configuration
- Performed multi-variable **optimization** to balance performance, weight, and cost of the aircraft structure.

SAE Aero Design

Chief Engineer / Project Manager

West Lafayette, IN

May 2025 – Present

- Spearheaded a wind-tunnel test campaign to analyze control surface authorities and drag prediction.
- Implemented a lessons-learned database from **5+** past flight failures to eliminate recurring design errors.
- Authored a **75-page** design guide and created a standardize onboarding to reduce training time for members.

Purdue Space Program – A SEDS Chapter

Satellites Deputy Systems Director

West Lafayette, IN

Feb 2024 – Nov 2024

- Improved project progress with **~15** engineers and other sub-teams ensuring alignment with project goals.
- Formulated, reviewed, refined **200+** project and system-level requirements for Boiler Bus (**~\$100K budget**).
- Manufactured** the satellite frame end plates using **CAM** and **4** wall types through **laser cutting**.

Purdue Aerial Robotics Team

Competition Airframe

West Lafayette, IN

Oct 2023 – May 2024

- Performed analysis using XFLR5 to select optimal airfoils, improving lift-to-drag efficiency by **~10%**.
- Engineered aircraft parts for the AUVSI SUAS competition through **NX** and Teamcenter with **~30** engineers.
- Applied engineering solutions to resolve manufacturing issues of molds, carbon fiber, and internal structures.

RESEARCH PUBLICATIONS

Authored "Characterizing Risk from Propulsion System Failures in Terminal Operations in UAMs" for AAE 571